Bicycle and Pedestrian Plan for Eakins Oval and Fairmount, Philadelphia

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I. Introduction. Existing resources, current conditions

（1）Site location

About site location, the site is located in central Philadelphia, on the southwest side of Central City. Bordered by the Schuylkill River to the west, Spring Garden St to the north, Pennsylvania Ave to the east, and Benjamin Franklin Pkwy to the south.



（2）History

The Benjamin Franklin Parkway is the premier example of a City Beautiful-era boulevard in the United States, and also a 105-year old continual project. Simply put, it is the physical armature of the connection from the heart of Penn's original gridded city to the beginning of Fairmount Park, a way of transitioning urbanity to forested park, marked with major cultural institutions like the Franklin Institute and Museum of Art. As the decades progressed, however, and the overwhelming dominance of car travel became apparent, the Ben Franklin Parkway too turned from a pleasant pleasure drive to a conduit for Philadelphia's major arterial parkways.

From City Hall, the Parkway cuts through William Penn's original grid as it reaches its terminus at 24th Street. Here, the Parkway turns into a massive traffic rotary circle, Eakins Oval, where it collects and distributes traffic from three major arterials: Spring Garden Street East and West, Kelly Drive, and Martin Luther King Drive. A product of a circa-1960s redesign by head city planner Ed Bacon, the "Oval" extends major roadway slip lanes and overpasses for the drives, resulting in sustained 40+mph car traffic speeds through complex merges and weaves. At its widest point, going westbound between Spring Garden St, it is five 12-foot car lanes and a buffered bike lane wide; no section is less than three lanes wide.

Fundamentally, as with the rest of the Parkway, the entire complex of Eakins Oval is **designed solely around cars**, with cyclists and pedestrians viewed as an afterthought. It is hoped that this plan, along with other promising initiatives such as those produced by the City in the past year, will help to completely decentralize and deemphasize cars on the Parkway in favor of cyclists and pedestrians, and restore the original intent of the area as a recreational park in a vibrant cultural neighborhood.



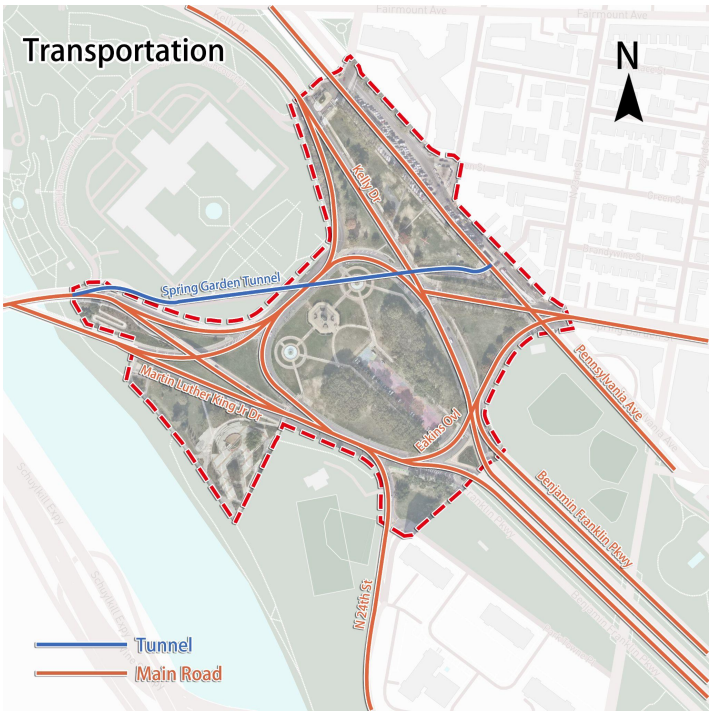


Eakins Oval and the Parkway do not simply exist as a grade separated highway either; as mentioned with the slip lanes, there is a complex network of feeder and neighborhood roads that flow to and from the site, none of which adhere to the overarching grid of Center City. A mess of one-way slip lanes and acute-angled intersections results, creating an unpredictable and awkward environment for cyclists and pedestrians who often have insufficient time to cross abnormally wide roads.

Starting from the north, Kelly Drive runs along the eastern bank of the Schuylkill River as a four-lane high-speed arterial, with the inbound direction widening to three lanes before Eakins Oval. In the AM peak, Kelly Drive southbound contributes the highest volume of vehicles to the Oval, which cements its status as the main continuation of Parkway traffic. Especially southbound, Kelly Drive traffic is so dominant that there is infamously no direct crosswalk from the Art Museum's Rocky Steps to Eakins Oval

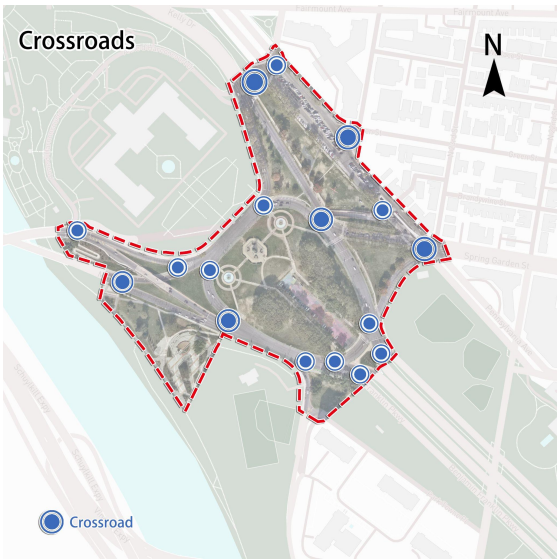
（3）Transportation

Regarding the traffic on the site, the roads within the site are mainly urban arterial roads, most of which have four lanes and a few have two lanes, and almost all of which are one-way. In addition, the site includes a tunnel and a traffic oval. The roads here are of good quality and are well used. There is also little shade on the roads, providing good visibility for pedestrians, cyclists, and motorists.



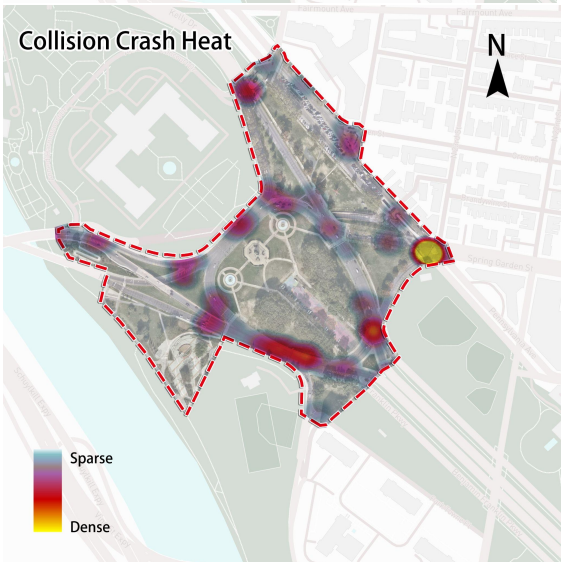
(4) Activity point-Green space-East Coast Greenway

With the aforementioned context, the primary benefit of improving the Eakins Oval area is self-evident: linking Center City's two major bike arteries together as part of the East Coast Greenway



(5) Crossroad

There are 16 intersections on the site, with six intersections with two lanes of traffic and ten intersections with more than two lanes of traffic. The intersections are mainly around the traffic circle, with five intersections on the west side of the traffic circle where it connects with Benjamin Franklin Pkwy.



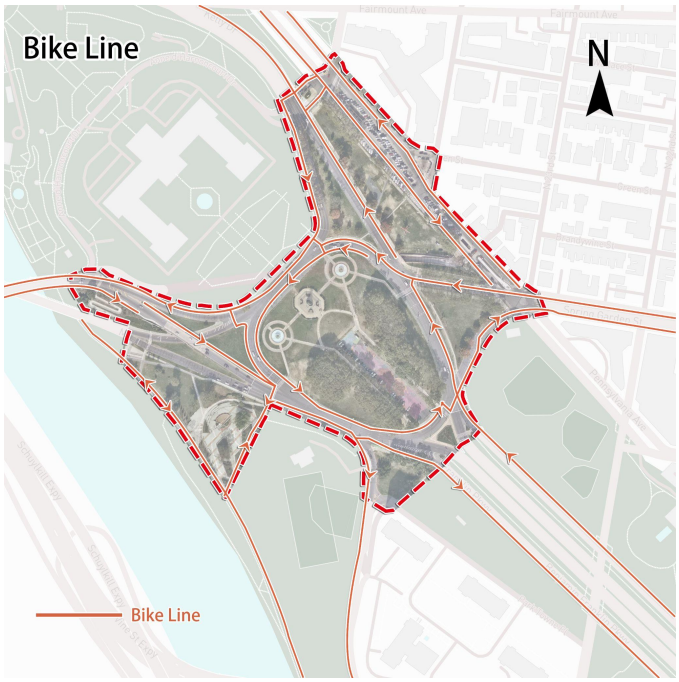
(6) Crash

Regarding the collision crash heat map in the site, it can be seen from the figure that the main collision locations in the site are intersections. As the intersection on the east side is more complicated, the number of collisions is also the most frequent. The next place where traffic accidents are likely to occur is the bike path on the southwest side of the site, where people are prone to

traffic accidents due to the lack of reasonable traffic signal placement. In addition, the entrance of the art museum is also a place where collisions are likely to occur due to the high flow of people.

（7）Bike line

The existing bike lanes on the site are relatively well developed, with bike lanes mainly arranged around the traffic circle and dispersed outward from the traffic circle. The bike lane on the west side connects to the park, the south side connects to Benjamin Franklin Pkwy and the east side connects to the center city. The bike lanes on the road within the site are all one-way lanes and the lanes in the park are two-way lanes.



（8）Entrance and vitality point

The main activity spaces within the site are in the traffic circle in the middle of the site and in the park on the west side of the site, where the crowd is more concentrated. Regarding the entry of the site, most pedestrians will enter the site from the art museum on the north side, the residential area on the east side, and the park on the east side. Since the south side of the site is connected to Benjamin Franklin Pkwy and the west side is connected to Spring Garden St, most of the vehicles will enter the site from these two sides.



II. Data Analysis

To analyze the biking and walking demands and socio-economic background on the site, we collect travel and demographic data from the US Census Bureau's American Community Survey (ACS).

(1) Demographic composition in the study area

According to the demographic information from 2019 ACS 5-year data, the study area has a high diversity population with 62% of white population, 20.6% of black or African population, and 12.2% of Asian population. 16% of the population is below poverty. In terms of car ownership, 21% of the households don't have vehicles, 50% of the households there have one vehicle, and about 28% of the households have more than one vehicle.

No vehicle	One vehicle	Two vehicles	Three vehicles	Four vehicles	Five or more vehicles
21%	50%	26%	3%	1%	0%

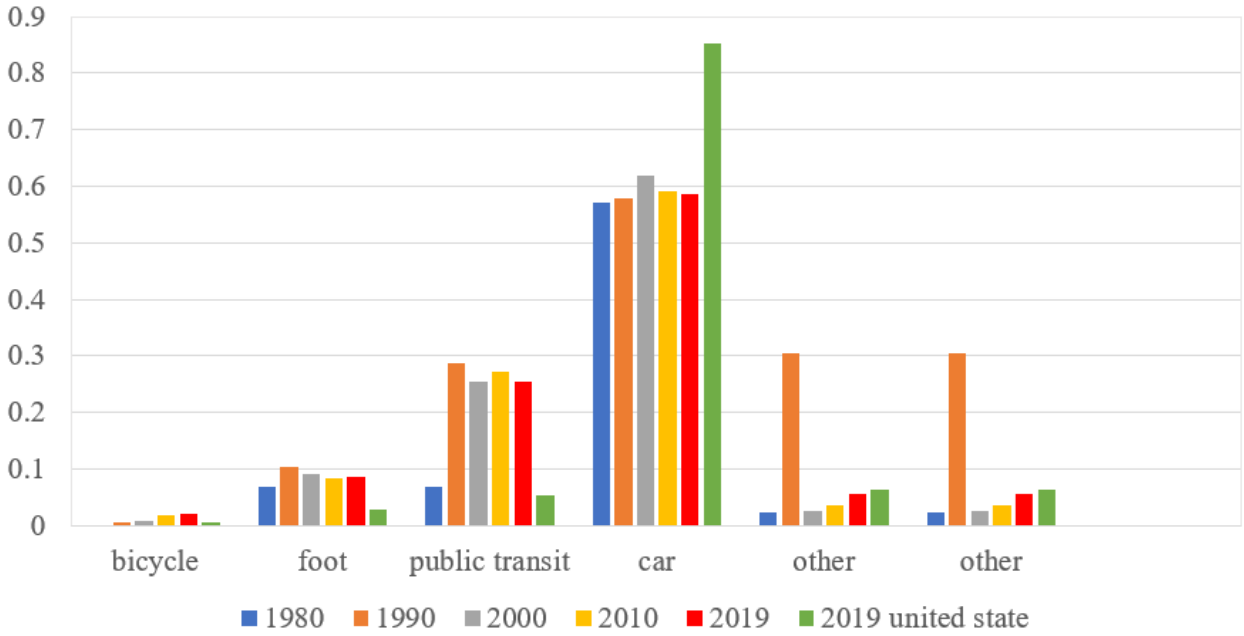
Table1: vehicle ownership in the study area

(2) Transportation mode share in Philadelphia and study area

According to the commuting information from ACS 5-year data, bicycling and walking are in high demand for people in Philadelphia. 8.5% percent of people walk to work, and bicycle transportation mode increased from 0.6% in 1990 to 2.1% in 2021, while the average of transportation modes in walking and bicycle in the United State in 2019 are 2.7% and 0.6% correspondingly.

	bicycle	foot	public transit	car	other
1980	N/A	6.90%	6.90%	57.10%	2.20%
1990	0.60%	10.40%	28.70%	57.80%	30.50%
2000	0.90%	9.10%	25.40%	62.00%	2.50%
2010	1.80%	8.30%	27.20%	59.10%	3.50%
2019	2.10%	8.5%	25.4%	58.5%	5.50%
2019 united state	0.6%	2.70%	5.2%	85.3%	6.20%

Table2: Transportation mode in Philadelphia from ACS 5-year data



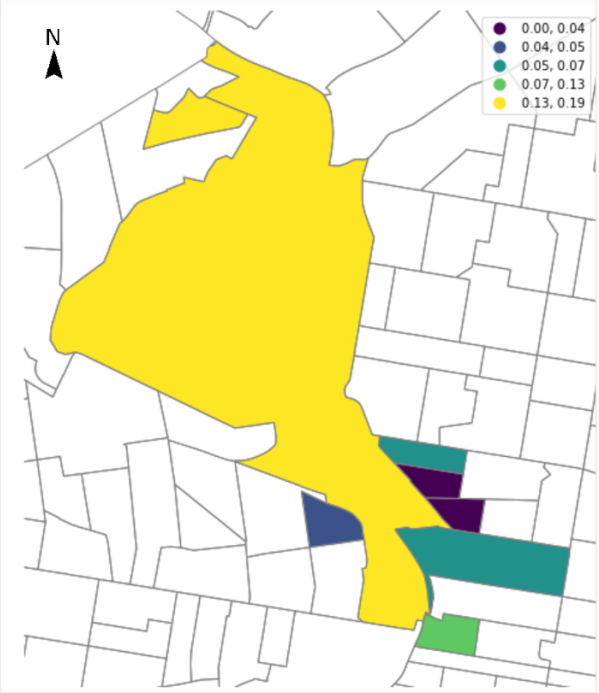
Transportation mode in Philadelphia from ACS 5 year data

In addition, we analyzed the transportation mode on the site based on tract level from ACS 2019 5-year data. Compared to the transportation mode in Philadelphia, the bike mode in the study area is 138% higher than the average level and arrives at 5%; the foot mode is 182% higher than the city's average value, and arrives at 24%.

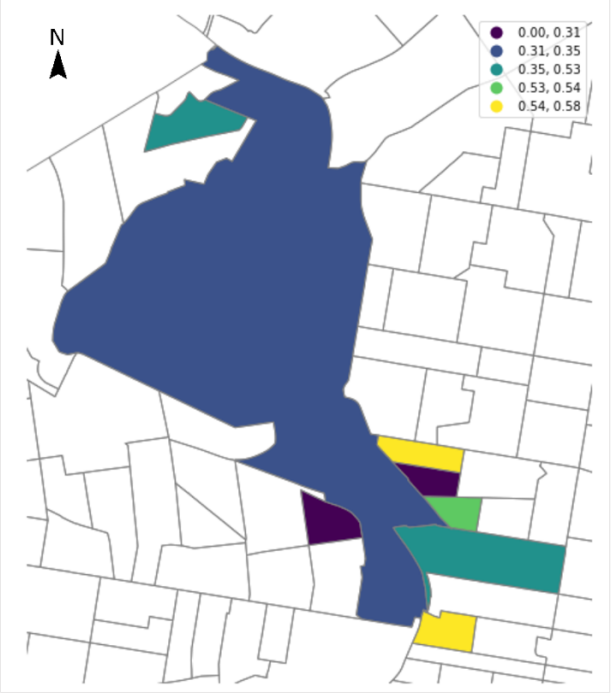
Walk	Bike	Private Vechicle	Public Transit	Other
24%	5%	45%	16%	10%

Table3 : Transportation mode in study area from ACS 2019 5-year data

Combined with geojson data from Tiger web survey, we collect geometry boundaries of tracts and merge them with ACS 2019 5-year data to make a choropleth map which displays the spatial distribution of bike mode and walk mode in the study area. According to the map below, we can clearly see most areas on the site have a high proportion of bicycle transportation mode sharing, while have relatively low proportion of walking rate, which are around Fairmount Park along the Schuylkill River.



Bike mode sharing in study area

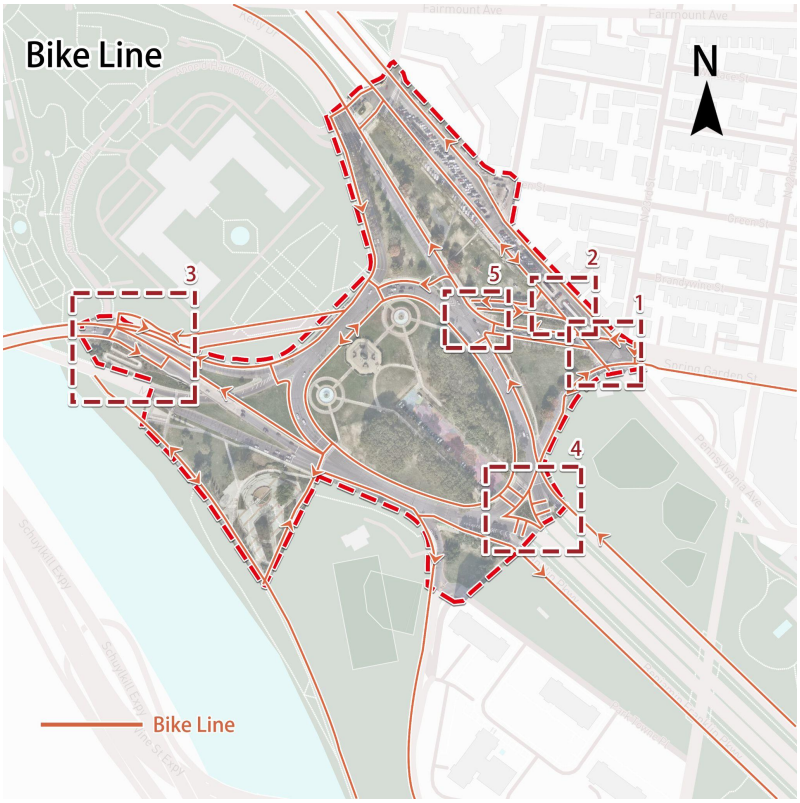


Walk mode sharing in study area

To sum up, we concluded that with the diverse population, bicycling and walking are in high demand in the study area.

III. Proposed improvement

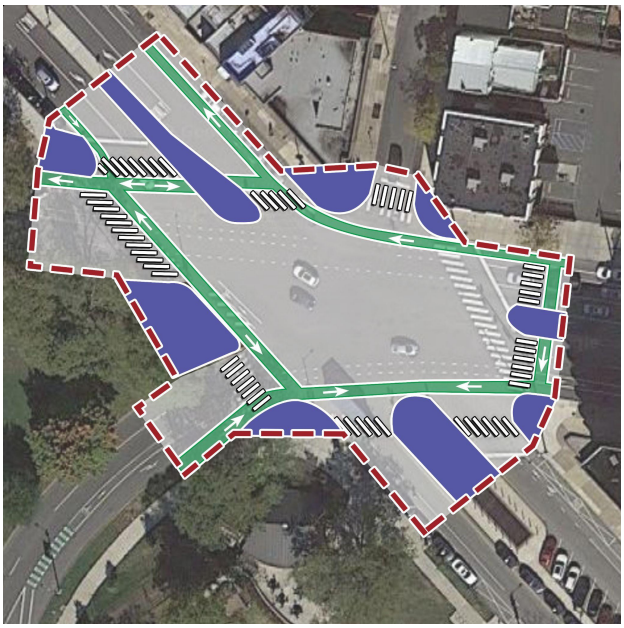
Based on the previous analysis and on-site investigation, the connection of pedestrians in this area is relatively good, so we will mainly focus on bike lanes. Overall, we will give all sidewalks ADA ramps and give not only sidewalks but also bike lanes the protection they need. More importantly, bikes and pedestrians will receive a dedicated walk phase or leading pedestrian interval, which is an urgent demand in terms of the safety of cyclists and peds. For the bike lanes in the whole area, we plan to open the Spring Garden Tunnel only for cyclists. Currently, it is only for vehicles from east to west and it's full of trash, like a



half-closed tunnel. Once opened, the connection from the other side of the river or the center city to the Spring Garden neighborhood will be greatly improved, and the whole path will be very smooth and straightforward. Another big change is that we plan to add a cycle path around the green area of Eakins Oval, because we found that many cyclists would not ride on the outer side according to the existing bicycle lanes, and many of them will pass directly through the middle of the green space. Accordingly, we made huge changes to five intersections. The adjustment and images are as follows.

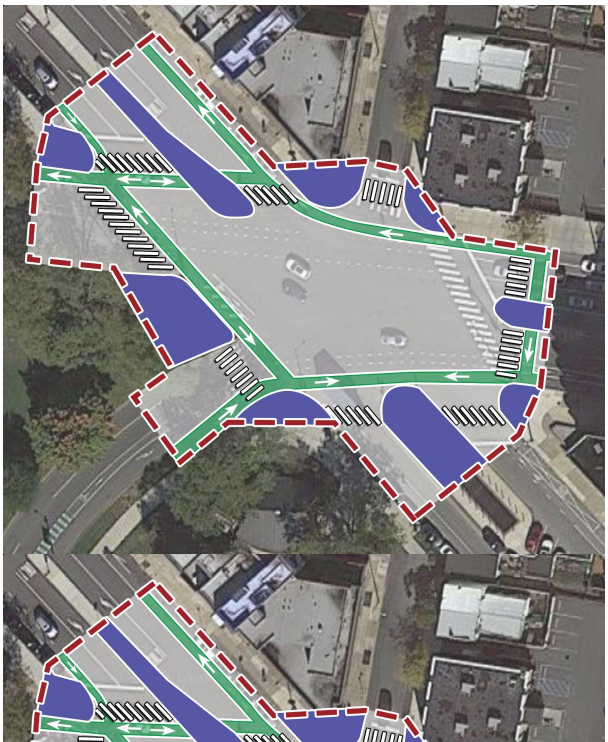
（1）Intersection 1

The main problem at intersection 1 is the connection of bicycle lanes and the excessively large intersections, which causes pedestrians and cyclists to cross the road for too long, meaning the potential risk is high. So, we plan to build an isolation island on Spring Garden St on the west side of the intersection, and change the sloping zebra crossing to be perpendicular to the road. In addition, in order to cooperate with the opening of Spring Garden Tunnel, we have differentiated the direction of bicycle lanes at this intersection. All directions are two-way, except the path from Spring Garden St to Pennsylvania Ave, which is one-way.



（2）Intersection 2

Intersection 2 is one of the entrances and exits of Spring Garden Tunnel, and its main adjustment is the change of bicycle lanes. Other things remaining unchanged, Pennsylvania Ave will be converted the one-way bike lane from northwest to southeast to bi-directional path to accommodate tunnel access. Considering the potential risk of conflict, we will strictly distinguish the specific directions and give clear instructions for the dual bicycle lanes at this entrance and exit.

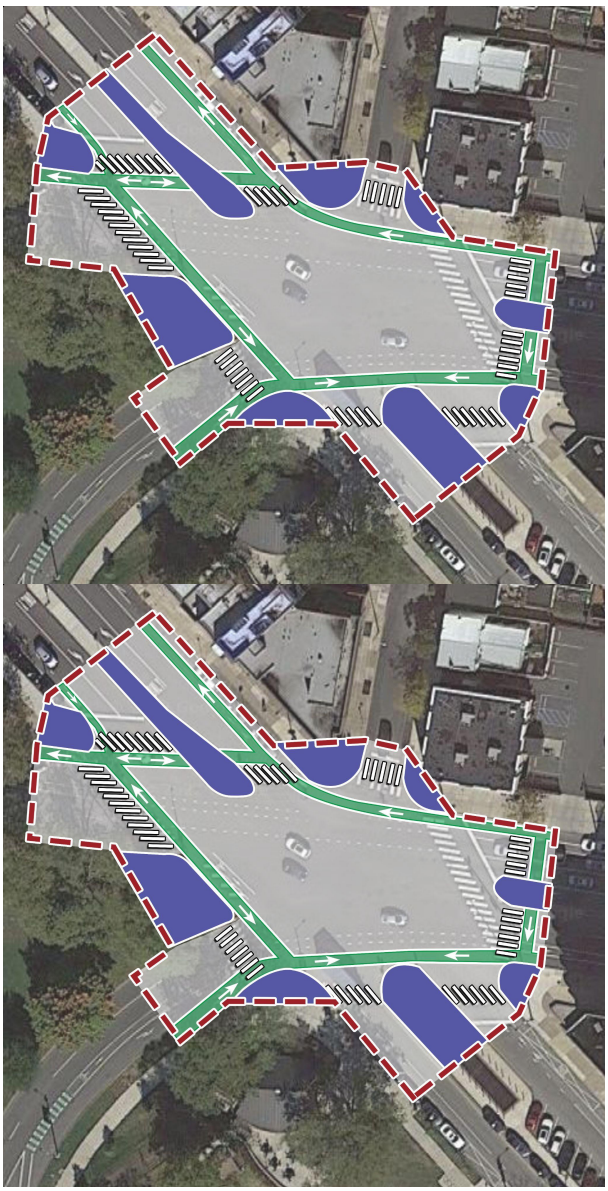


(3) Intersection 3

Intersection 3 is another entrance and exits of the Spring Garden Tunnel. Because the tunnel currently only allows passing of vehicles from east to west, the general situation of this intersection conforms to our design. The change that needs to be made is a separate green light time for cyclists and pedestrians. After setting up the traffic light program, the only thing to do is to draw the well-designed and fully-connected bicycle lanes according to the graph.

(4) Intersection 4

Intersection 4 is an improvement to match the circular bike path. Our bike lanes and crosswalks were rerouted to the middle of the arc, and a traffic light was added to the extreme south of the entire circular green space, which controlled cyclists and pedestrians crossing Eakins Oval and Benjamin Franklin Pkwy. The original stop line for vehicles on the Eakins Oval will be moved back before the fork between the Eakins Oval and Benjamin Franklin Pkwy.



(5) Intersection 5

Intersection 5 is a high accident site in our previous analysis, because the bike lanes are not connected to each other, resulting in different cycling routes for different cyclists. And the fact that vehicles from Benjamin Franklin Pkwy to Kelly Dr cross with bicycles from Benjamin Franklin Pkwy to Eakins Oval is a major cause of accidents. Therefore, we strictly stipulate the bicycle lanes as shown in the figure and we move crosswalks back from this intersection. Although it will spare the cyclists a little distance, this will undoubtedly greatly ensure the safety of cyclists and pedestrians

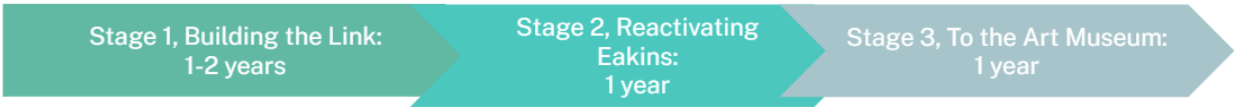
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IV. Plan implementation

This chapter outlines the overall strategy, approach and process for implementing the pedestrian and bicycle improvements of the plan. The basic strategy we developed was to implement road changes in subdivided steps without affecting commuting patterns in the study area. People will unknowingly discover more

convenient and safer options, which can help promote wider recognition and acceptance of new transportation routes. After construction, the new transportation network in front of the Philadelphia Art Museum will further increase the effectiveness of the city's pedestrian and bicycle transportation system.

Pedestrian and bicycle improvements in Philadelphia are often not done as stand-alone projects, but as part of larger road and streetscape improvement projects, which makes it difficult to develop precise phased strategies for planning proposals, the purpose of this project is purely. Therefore, in this ideal case, we will give a more detailed phased strategy. Within the strategy, flexibility and economy are also taken into consideration, in order to meet the comprehensiveness and achievability of the implementation process. According to the RAISE Grant eligible project, the cost of a transportation infrastructure project is in the range of $10,000,000 to $15,000,000. Based on the size of the different tasks in our overall project, we have made a rough division of the overall funding.



（1）Stage 1

We plan to implement our plan on the intersections 1, 2, 3 and the tunnel first, which will last for 1-2 years in our expectation, because this is the most critical and most prominent road change in the entire area. Once completed, it will greatly increase the comfort and convenience of cyclists and at the same time greatly reduce the occurrence of crashes. We will allocate $2,000,000 for the opening of the Spring Garden Tunnel, $3,000,000 for the reconstruction of intersection 1, $2,500,000 for the change of the second intersection and 1 million for the adjustment of the third intersection.

The entire Spring Garden Tunnel is currently only for vehicles from Pennsylvania Ave to Spring Garden St from east to west, and the first thing we need to do is to close and block the tunnel. Since the current tunnel is in disrepair and full of garbage, it is far from the standard of opening directly, as a result the reinforcement and cleaning of the tunnel is the primary task.

The changes at the No. 2 and No. 3 intersections are mainly attached with the opening of the Spring Garden Tunnel, so it needs to be done at the same time as the tunnel dredging. We hope that the parallel process can be carried out under the condition of sufficient manpower, because this will shorten the construction period and reduce the time cost, but if the conditions cannot be met, it can be processed centrally after the tunnel is built. And if there is a situation where there is no such enough manpower, just as the second situation we mentioned, there will also be a sequential relationship between No. 2 and No. 3. Since the reconstruction of intersection 3 involves several important lanes and requires changes to traffic lights, it will affect commuting

not only for bicycles, but also for motor vehicles. At the same time, since intersection 3 is the only way to cross the river from the city center to the Spring Garden area, we need to take care of it first. The main changes at intersection 3 are changes to bike lanes and sidewalks, as well as changes to traffic lights. Because we designed the green-light-time for cyclists to pass, the built-in program for the traffic lights at this intersection needed to be replaced. We recommend changing the traffic lights first, and then changing the lanes, because the current road conditions at the intersection can meet the needs of the new traffic light program, and the unreasonable bicycle lanes can be reduced by quickly drawing new road lines to reduce potential risks. The major changes to Intersection 2 are the redesign of the bike lanes on Pennsylvania Ave from the northwest to the southeast. Because this only involves a change in one direction and a small distance, it can be completed easily and quickly. And that's the reason why we recommend that the construction of Intersection 2 be completed after the tunnel is cleared and Intersection 3 is built.

The final highlight of the first stage will come on stage, that is, the reconstruction of Intersection 1. Because No. 1 Intersection is the same as Intersection 3 as the other end of the only entrance and exit from the city center to the Spring Garden area, its importance is obvious. In addition, there are as many as 6 roads to and from this intersection, and 7 pedestrian crossings at present. These numbers undoubtedly show its significant position in the whole area. Through history, we can find that this intersection has always been the focus of contradictions for planners. Due to its complexity, this intersection has been modified many times. Therefore, how to promote our change simply and effectively is what we need to consider urgently. There are two main changes, one is the construction of the isolated island in the middle of the east of Spring Garden St, and the other is painting the new bicycle lanes and crossings. Our recommendation is to build islands first, as it will result in fewer vehicle lanes. While there may be resistance from some vehicle enthusiasts, this is the best way we can think of to shorten pedestrians and cyclists crossing the street. After the island is built, we can draw the bike lanes and sidewalks on the road based on the blueprints. Different bike lanes have different directions, and we recommend to draw one-way lines first, like the bike path from Spring Garden St to Pennsylvania Ave, and then two-way lanes, like a bike lane going northwest to southeast along Pennsylvania Ave. Because the bicycle lanes at this intersection are currently disconnected, and the direction is unknown, constructors must pay attention to filling and marking during the drawing process.

(2) Stage 2

In our plan, stage 2 will take one year. In stage 2, We plan to implement the bike lanes redevelopment at Eakins Oval around the green space and the construction of intersection 4. Once Phase 2 is completed, the overall appearance of the area will be completely new and the safety of cyclists and riders will be greatly guaranteed. We plan to spend $1,500,000 for the

construction of the bike lane around Eakins Oval and $1,000,000 for the construction of intersection 4.

We learned that the reason for not installing traffic lights in the west-to-east part of Eakins Oval is the fear of congestion, but because of the feature of multiple exits, the bicycle lanes located on the outer lanes face a huge threat from cars changing lanes to the exits, so we decided to put the bicycle path on the inner road. In this case, the construction of this circular bicycle path will not affect the commuting of any vehicles or pedestrians. Therefore, we recommend building this part of the bike path first. Once completed, we'll move on to intersection 4 to guide cyclists and pedestrians using the new bike lanes and sidewalks. We will install a traffic light at the southernmost point of the entire circular green area, which controls cyclists and pedestrians crossing Eakins Oval and Benjamin Franklin Pkwy. It's just a small traffic light that doesn't affect the layout of the road or green space, so it's easy to install. The drawing of bicycle lanes and zebra crossings is also a very simple job, it will not occupy the lane for a long time, and commuting can be guaranteed.

（3）Stage 3

We completed most of the major reshaping of the front area of Philadelphia Museum of Art in the first two phases within 2-3 years, and the final phase was tasked with completing some minor intersections, including a large one, intersection 5. We plan to spend one year on the final stage and $500,000 for each task.

Frankly speaking, intersection 5 cannot be counted as an intersection, but just as a junction of vehicle entrances and exits. In our previous analysis, we found that this place is also in urgent need of improvement. Even so, the transformation of this fork will be a simple job of connecting the broken bike lanes. Because our work is done with cyclists and pedestrians, vehicle-specific behavior planning is out of our consideration. However, in order to ensure overall commuting, we still recommend drawing the bicycle lanes along the road first, and then the bicycle lanes and zebra crossings that cross the road.

The hardest mission in this phase should be the overpass over the exit from Eakins Oval to North 24th St. This flyover is the safest way we can think of connecting Paine's Park with the main road. It is believed that after the completion of the overpass, pedestrians and cyclists will like and enjoy such a way of crossing the street.

V. Evaluation metrics and progress

Aiming at improving pedestrian and bicycle safety, commuting efficiency and transportation mode sharing, we assess our proposal from qualitative and quantitative perspectives. The

measurement tools consist of 1) Surveys to measure the transportation sharing mode and bike stress level; 2) Observations to measure the actual road condition; 3) Interviews with pedestrians and cyclists to measure their perceptions of walking/biking infrastures; 4) Benefit Analysis.

（1）Surveys

Firstly, data related to people's travel behavior are important to assess travel demand and supply. Open data from ACS 1-year estimates are useful to get information such as transportation mode sharing, travel activity as well as demographic characteristics. Household Travel Surveys are another kind of data which provides the accurate information related to households' transportation mode choice, as well as the frequency, length and purpose of trips. Also, bicycles and pedestrians count data from DVRPS based on non-motorized monitoring programs can be important to detect the usage of transportation infractures .

Secondly, data related to road safety and fatality rate needs to be included in the assessment. DVRPS's bike level of traffic stress(LTS) is based on the number of lanes, effective vehicle speed and the presence and type of bicycle facility based on road segment, which provides people's mental evaluation of the bicycle environment(DVRPC,2022). Vehicular crash data within pedestrians and cyclists from OpenDataPhilly provides objective evaluation for road safety.

（2）Observations

Follow-up observations to study people's natural transportation behaviors are important. Observing if people obey the transportation rules with our facility design can tell if our plan meets people's commuting needs.

（3）Interviews

To gain feedback on the effectiveness of the plan and implementation process, interviews with transportation attendees are necessary. In the interviews, we are going to survey people's perceptions of barriers to walking and biking such as car traffic, lack of crosswalks and their attitudes to plan implementation effects.

（4）Quantitative benefit Analysis

Combined with the data collected above, a quantitative assessment of the economic benefits of the implementation plan can be illustrated with a focus on congestion, climate change and physical activity(Fishman, 2011). The congestion benefits are determined by decreased vehicle kilometers by walking and biking at peak hour due to plan implementation. The climate changes are calculated using the decreased number of other transportation means with corresponding environmental costs. The physical activity benefits equal to the travel kilometers by increased bike and pedestrians multiply the monetized health benefits of cycling/walking per kilometer(Arsenio, 2015).

Input
Number of bicycles
Trips per day per transportation means
Percentage of trips per transportation means in peak hour
Transportation mode sharing
Travel distance per day
Health benefits for walking/km
Health benefits for cycling/km
Congestion reduction benefit/km
Climate change benefits/km
Output
Congestion benefit per year
Climate benefit per year
Physical activity benefit per year

The pipeline of the quantitative benefit Analysis

Literature

[1] https://www.dvrpc.org/traffic/bikepedtravelmonitoring
[2] https://www.opendataphilly.org/dataset/vehicular-crash-data
[3] https://www.dvrpc.org/webmaps/bike-lts/
[4] Fishman, Elliot, Simon Washington, and Narelle L. Haworth. "An evaluation framework for assessing the impact of public bicycle share schemes." Transportation Research Board (TRB) 91st Annual Meeting. 2012.
[5] Arsenio, Elisabete, and Paulo Ribeiro. "The economic assessment of health benefits of active transport." Sustainable Urban Transport. Emerald Group Publishing Limited, 2015.
[6] https://www.transportation.gov/RAISEgrants